Classical histories, complete records and the cusp limit

Result, 2026-10-28; written derivations, unrefereed. For the bounded Newton cusp already used in the preparation-floor note:

1. There is a classical, deterministic, time-reversal-symmetric group on the space of complete solutions. Its instantaneous (q, p) projection follows Newton's equation, and it admits exact states and ordinary selective records with zero covariance action. The group cannot descend to an evolution on instantaneous phase-space probabilities.
2. For Gaussian classical preparations whose dimensionless width is ϵ, the initial covariance action is $S_0\epsilon^2$. All their initial laws tend to the same point as $\epsilon \to 0$, while the future law can tend to any prescribed mixture of the two departing trajectories. At each fixed $\epsilon > 0$ even complete-trajectory statistics depend continuously on the Gaussian center; the modulus is not uniform in ϵ.
3. Closure under arbitrarily fine, nondisturbing joint (q, p) records forces the covariance-action infimum to zero, without any weak-closure or canonical-covariance assumption. Thus this record postulate cannot be added to the three hypotheses of the cusp preparation-floor theorem.
4. A separate continuity-and-copying theorem forces a positive covariance action without assuming reversibility, provided evolution actually transports the admitted Gaussian states by the cusp law. A classical entropy-restricted model meets those hypotheses with a supplied scale.

This is a test of the missing premise, not a derivation of $\hbar > 0$. The trajectory model has extra, generally future-selecting classical state information and no constructed symplectic or local apparatus structure on that enlarged space. A probability of selecting a branch is not quantum coherence. The Gaussian width ϵ is a dimensionless *classical preparation parameter*, not Planck's constant. The quantum comparison in section 6 retains the physical $\hbar$ explicitly.

The route has substantial prior art. Sell 1973 (publisher abstract read) treats dynamical systems for nonunique ODEs; the solution-space group below is given its own proof. The branching of vanishing classical uncertainty is an instance of the established subject of spontaneous stochasticity: see Eyink and Drivas 2015 (author abstract read) for the related rough-potential quantum and classical context, and Vanpoucke and Wenmackers 2021 (author abstract read) for probability assignments in the same power-law non-Lipschitz mechanical setting. No literature priority is claimed for the construction or the Gaussian weights here. The repository's two-pointer theorem already shows operational failure of a prior density ceiling under specific Hamiltonian instruments; section 5 gives a more general, explicitly conditional version at the probability-law level.

1. Force and notation

Fix $m, \gamma, R > 0$, $0 < \alpha < 1$, and a smooth even cutoff χ with $0 \le \chi \le 1$, $\chi = 1$ on $[-1, 1]$ and $\chi = 0$ outside $[-2, 2]$. Consider

$$V(q) = -\frac{\gamma}{1+\alpha}|q|^{1+\alpha}\chi(q/R), \qquad \dot{q} = p/m, \qquad \dot{p} = -V'(q). \tag{1}$$

The force is continuous and bounded, although not Lipschitz at the origin. For every initial (q, p) there is a global solution; the proof and the uniform bounds used below are in the cusp preparation-floor note, §1. Put

$$k = 1 + \alpha, \quad \lambda = \frac{\gamma}{1+\alpha}, \quad \beta = \frac{2}{1-\alpha}, \quad B = \left[\frac{\gamma(1-\alpha)^2}{2m(1+\alpha)}\right]^{1/(1-\alpha)}. \tag{2}$$

For $0 \le t \le T$ with $BT^\beta < R$, the two rest-departure solutions are

$$\Gamma_\pm(t) = \big(\pm Bt^\beta,\ \pm m\beta B t^{\beta-1}\big). \tag{3}$$

Let $L_0, P_0 > 0$ be arbitrary calibration units, $S_0 = L_0 P_0$, and write $z = (q/L_0, p/P_0)$. For a finite-covariance law ν put $\kappa(\nu) = \sqrt{\det \mathrm{Cov}_\nu(q,p)}$. The calibrations do not introduce a physical lower bound.

2. The trajectory-state completion

Let Ω be the set of all globally defined solutions of (1), viewed as functions $\gamma : \mathbb{R} \to \mathbb{R}^2$ with the topology of uniform convergence on compact intervals. Write $e_t(\gamma) = \gamma(t)$ and

$$(T_s\gamma)(t) = \gamma(t+s), \qquad (\Theta\gamma)(t) = \big(q(-t), -p(-t)\big). \tag{4}$$

Theorem 1. Ω is a nonempty Polish space. The maps T_s form a jointly continuous group of homeomorphisms and Θ is a continuous involution satisfying $\Theta T_s \Theta = T_{-s}$. Every e_0-fiber is nonempty and compact. The state class $\mathcal{P}(\Omega)$ of all Borel probabilities has weakly continuous reversible evolution $\mu \mapsto (T_s)_*\mu$, contains point states, and supports ordinary classical conditioning on any Borel trajectory record of positive probability. Its instantaneous phase laws comprise all Borel probabilities on $\mathbb{R}^2$; in particular their covariance-action infimum is zero.

Proof. The integral form of (1) is preserved by local-uniform limits: on each compact time interval the convergent paths have a common compact range, and the continuous vector field is uniformly continuous there. Thus Ω is closed in the Polish compact-open space $C(\mathbb{R}, \mathbb{R}^2)$. Time translations and time reversal preserve the autonomous equation and have the asserted algebra and continuity.

Given a compact set of initial points, the bounded-force estimates in the cusp note give locally uniform bounds on q, p and their derivatives. Arzelà–Ascoli, a diagonal subsequence and closedness give compactness of $e_0^{-1}(K)$. Global existence makes every fiber nonempty. Pushforward by the continuous time shifts gives the group on $\mathcal{P}(\Omega)$; for each bounded continuous test function, dominated convergence proves weak continuity in time. Ordinary restriction and normalization give the stated conditioning rule.

For the marginal claim, the proper continuous surjection e_0 has a Borel section $s : \mathbb{R}^2 \to \Omega$. One justification is the Kuratowski–Ryll-Nardzewski measurable-selection theorem (statement, theorem statement read): the compact-valued fiber map is Borel-measurable, since the image under a proper map of a closed set is closed and an open set in a metric space is a countable union of closed sets. Hence $s_*\nu$ is a history preparation with instantaneous marginal ν for any $\nu \in \mathcal{P}(\mathbb{R}^2)$. A Dirac history has a Dirac instantaneous marginal, with $\kappa = 0$. □

The theorem constructs reversible *history* evolution, not a reversible map on laws of (q,p) alone. Take two histories that arrive at the cusp from opposite sides. At their common arrival time their instantaneous marginals are both $\delta_{(0,0)}$, while the full states and their pasts remain distinct. The resting history and a departing history have the same instantaneous marginal at time zero but different future marginals, so no single map on phase laws can represent the shift for all history states. Moreover, the resting solution and the two solutions which leave after any prescribed delay $a \ge 0$ have the same entire (q,p) past up to time zero but different futures. Thus a deterministic completion representing all these possibilities must carry information beyond that past. In (4) the chosen future is part of the state. The construction does not show that this information has a finite-dimensional, local Hamiltonian realization or is accessible to a physical observer.

Because the phase marginal class is all of $\mathcal{P}(\mathbb{R}^2)$, it is weakly closed, affine-canonically invariant, and closed under classical product preparations, canonical mixing and marginalization **as a class of single-time laws**. The Borel section is a state-preparation selection; it is not a proof of a reversible canonical mixer acting on the enlarged history states. Nor does the

instantaneous marginal determine its later law. The preparation-floor theorem explicitly assumes that a reversible evolution acts on those marginals themselves; Theorem 1 does not satisfy that premise, so the two results are consistent.

3. A continuous classical family with nonuniform zero-width limit

Let X, Y be independent standard normal variables. For fixed $u \in \mathbb{R}$ and $\epsilon > 0$, prepare

$$q_0 = \epsilon L_0(u + X), \qquad p_0 = \epsilon P_0 Y. \tag{5}$$

The energy-zero set is Lebesgue-null, so (5) selects unique global trajectories almost surely. Equivalently, lift the law by the section s: different choices of s agree almost surely here. Let Φ denote the standard normal distribution function.

Theorem 2. Let $0 < T < (R/B)^{1/\beta}$. As $\epsilon \to 0$, the initial law of (5) converges weakly to $\delta_{(0,0)}$ for every fixed u, and

$$\kappa_\epsilon = S_0 \epsilon^2, \qquad \mathrm{Law}\big((q(t), p(t))_{0 \le t \le T}\big) \Longrightarrow \Phi(u)\, \delta_{\Gamma_+} + \Phi(-u)\, \delta_{\Gamma_-} \quad \text{in } C([0,T], \mathbb{R}^2). \tag{6}$$

Every mixture weight strictly between zero and one is obtained by a fixed u. For either pure branch one may instead take $u = u_\epsilon \to \pm\infty$ with $\epsilon |u_\epsilon| \to 0$. The distributions are classical ensembles of Newton trajectories, with no supplied $\hbar$.

Proof. The first limit and covariance are immediate. Fix a realization with $u + X \neq 0$ and finite Y. Its initial energy is, for small ϵ,

$$E_\epsilon = \frac{\epsilon^2 P_0^2 Y^2}{2m} - \lambda\, |\epsilon L_0(u + X)|^k < 0. \tag{7}$$

Indeed $k < 2$. Thus it cannot reach $q = 0$, so its sign remains that of $u + X$. Its turning distance $a_\epsilon = (-E_\epsilon/\lambda)^{1/k}$ satisfies $0 < a_\epsilon \le |q_0|$, and tends to zero. Inside $|q| < R$, separation of the energy equation gives the time from the turning point to $r \ge a$:

$$I_a(r) = \sqrt{\frac{m}{2\lambda}} \int_a^r \frac{dv}{\sqrt{v^k - a^k}}. \tag{8}$$

For fixed $0 < r < R$, $I_a(r) \to I_0(r)$ as $a \downarrow 0$, where

$$I_0(r) = \sqrt{\frac{m}{2\lambda}} \frac{2}{1-\alpha} r^{(1-\alpha)/2}, \qquad I_0^{-1}(t) = B t^\beta. \tag{9}$$

For the convergence, the part from a to $2a$ is $O(a^{(1-\alpha)/2})$; on $v \ge 2a$ the integrand is dominated by a constant times $v^{-k/2}$, which is integrable at zero. Moreover,

$$I_a(r) \le \sqrt{\frac{m}{2\lambda}} \frac{2}{1-\alpha} (r - a)^{(1-\alpha)/2}, \tag{10}$$

since $v^k - a^k \ge (v-a)^k$ for $k > 1$. Hence the time from the initial point to its turning point is $I_{a_\epsilon}(|q_0|) \to 0$. The outward time to any fixed r is $I_{a_\epsilon}(r)$ plus or minus this vanishing initial term. Bracketing fixed times by the hitting times of two nearby radii proves $q(t) \to \mathrm{sgn}(u+X) B t^\beta$ for each $t > 0$. The energy equation gives the corresponding momentum limit, with outward sign. For T as stated the limiting path remains inside $|q| < R$; the same hitting-time argument keeps the approximating paths there for all sufficiently small ϵ.

The bounded-force estimates give equicontinuity of both coordinates on $[0, T]$, so the pointwise convergence is uniform there. It holds for almost every (X, Y), and bounded convergence of continuous path functionals gives (6). If $u_\epsilon \to \pm\infty$ with $\epsilon |u_\epsilon| \to 0$, the same argument has eventual fixed sign for each (X, Y) and gives the corresponding pure branch. □

Fixed-width continuity. Let $\mu_{z,\epsilon} = s_*\mathcal{N}(z, \epsilon^2 I)$ on Ω, with z dimensionless. For all centers z, w and every t,

$$\left\|(T_t)_*\mu_{z,\epsilon} - (T_t)_*\mu_{w,\epsilon}\right\|_{\mathrm{TV}} \leq \frac{|z-w|}{\sqrt{2\pi}\,\epsilon}. \tag{11}$$

Here total variation is the supremum over measurable events. Pushforward cannot increase it. For the Gaussian bound, interpolate its center along the line from z to w and integrate the absolute directional derivative: half its integral is $1/(\sqrt{2\pi}\epsilon)$ per unit center distance. Equation (11) applies to any subsequent fixed measurable record, including a complete history record. Its constant diverges as $\epsilon \downarrow 0$.

The divergence cannot be removed uniformly. Fix $0 < t < T$ and a bounded continuous f with $f(Bt^\beta) = 1$ and $f(-Bt^\beta) = 0$; a clipped linear function suffices. For the two centers $z_\pm = (\pm\epsilon, 0)$, Theorem 2 gives

$$\lim_{\epsilon\downarrow 0}\left[\mathbb{E}_{z_+,\epsilon} f(q(t)) - \mathbb{E}_{z_-,\epsilon} f(q(t))\right] = 2\Phi(1) - 1 > 0. \tag{12}$$

The two initial centers are only 2ϵ apart. Thus no common modulus of continuity tending to zero can hold for these responses as ϵ varies. A uniform response modulus would exclude the zero-width sequence; it is an extra physical premise, not a consequence of continuity at each fixed width. More concretely, if the absolute response difference is bounded by $L|z-w|$ for some $L > 0$, uniformly over all allowed widths, and both translated preparations at each such width are admitted, (12) implies an $\epsilon_0 > 0$ such that widths below $\min\{\epsilon_0, [2\Phi(1)-1]/(4L)\}$ are excluded for this family. This conditional bound does not identify $S_0\epsilon^2$ with $\hbar$ and does not establish the response premise for physical records.

Corollary. No evolution on phase-space probability laws which agrees with the actual Newton transport of all the Gaussian laws (5) can be weakly continuous at $\delta_{(0,0)}$ at any fixed $t \in (0, T)$. Indeed all the inputs have that same weak limit while (6) gives distinct output limits for different u. This obstruction does not require injectivity or a rule for the exact point at the origin. It does require the indicated Gaussian preparations and their actual cusp transport.

4. A continuity selector without reversibility

The preceding corollary yields a different conditional action theorem. It uses actual Newton transport of nonpoint preparations, rather than compatibility only for point preparations if any exist.

Theorem 3. Use the W_2 metric in the calibrated coordinates, and let $\mathcal{S} \subset \mathcal{P}_2(\mathbb{R}^2)$ satisfy:

1. $\mathcal{S}$ contains a state with positive-definite covariance and is invariant under affine canonical transformations.
2. $\mathcal{S}$ is closed under W_2 convergence. Independent finite products of members, followed by copy-index orthogonal canonical mixing and retention of one output pair, produce members of $\mathcal{S}$, as specified in §1 of the central-limit note.
3. At one fixed $t \in (0, (R/B)^{1/\beta})$, there is a map $\mathcal{E}_t : \mathcal{S} \to \mathcal{P}(\mathbb{R}^2)$ which is continuous from W_2 convergence of inputs to weak convergence of outputs. For every nondegenerate Gaussian member of $\mathcal{S}$, its output is the actual almost-everywhere Newton transport under (1).

Then the infimum of κ on $\mathcal{S}$ is strictly positive. The map $\mathcal{E}_t$ need not be invertible, form a semigroup, or map back into $\mathcal{S}$ for this conclusion.

Proof. Work in calibrated coordinates. Suppose the infimum is zero and choose states with covariance matrices C_n and $\det C_n \to 0$. Translate their means to zero. Choose $\eta_n > 0$ such that $D_n = C_n + \eta_n I$ has $\det D_n \to 0$, and put $A_n = (\det D_n)^{1/4} D_n^{-1/2}$. In one pair $\det A_n = 1$, and

$$A_n C_n A_n^{\mathsf{T}} \le \sqrt{\det D_n}\, I. \tag{12a}$$

The transformed admissible states have second moment tending to zero, so they converge to δ_0 in W_2; closure admits δ_0.

By the central-limit result, a positive-definite covariance member yields an admissible Gaussian with the same covariance. Translation and a determinant-one transformation give an admissible $G = \mathcal{N}(0, \zeta I)$ for some $\zeta > 0$. For each n, take one independent copy of G and $n-1$ copies of δ_0, then use an orthogonal copy-index mixer whose first row is $(1, \dots, 1)/\sqrt{n}$. The retained pair has law $\mathcal{N}(0, (\zeta/n)I)$. Translations therefore admit both Gaussian inputs (5) with $\epsilon_n = \sqrt{\zeta/n}$ and $u = \pm 1$.

Both input sequences converge to δ_0 in W_2. The continuity assumption forces their output sequences to have the same weak limit $\mathcal{E}_t \delta_0$. Actual Newton transport and Theorem 2 give two different branch mixtures instead. This is a contradiction. □

The assumptions are consistent in a classical model. The entropy-floor class $\mathcal{E}_\zeta$ of §6 of the central-limit note has a **supplied** $\zeta > 0$, is W_2-closed, has the required canonical copy-index mixers, and is transported by the actual cusp flow. Its members are absolutely continuous. The classical flow is continuous at every initial state off the energy-zero set: bounded-force compactness and uniqueness there give continuous dependence. The exceptional set has Lebesgue measure zero, so the extended continuous-mapping theorem gives weak continuity of the transport on $\mathcal{E}_\zeta$ under W_2-convergent inputs. This provides a nonvacuous classical witness for the full continuity premise; its positive action is inherited from its imposed entropy bound.

The theorem therefore identifies a possible independently motivated stability requirement and makes its consequence explicit. It does not establish that all physical preparations obey the copy, closure and continuity rules, or that the selected floor is universal or quantum.

5. Fine records and the action floor

Theorem 4. Let $\mathcal{S}$ be a nonempty class of Borel probabilities on the classical phase plane. Suppose that, for every $\nu \in \mathcal{S}$ and every bounded open ball D in the calibrated $(q/L_0, p/P_0)$ plane with $\nu(D) > 0$, the ordinary conditional law $\nu(\cdot \mid D)$ is again in $\mathcal{S}$. Then

$$\inf_{\nu \in \mathcal{S},\ \operatorname{Cov} \nu < \infty} \sqrt{\det \operatorname{Cov}_\nu(q, p)} = 0. \tag{13}$$

If $\mathcal{S}$ is also weakly closed, it contains a Dirac measure.

Proof. Every probability has a point z_* in its support. The balls D_r of radius $r > 0$ centered there have positive probability. For the calibrated conditional covariance C_r, bounded support and centering at the conditional mean give $\operatorname{tr} C_r \le r^2$. Therefore

$$\kappa(\nu(\cdot \mid D_r)) = S_0 \sqrt{\det C_r} \le \tfrac{1}{2} S_0 \operatorname{tr} C_r \le \tfrac{1}{2} S_0 r^2. \tag{14}$$

These conditional laws have finite covariance and converge weakly to δ_{z_*} as $r \downarrow 0$. This proves both statements. □

The premise is an ideal, nondisturbing, arbitrarily fine *joint* record of classical position and momentum, with each positive-probability outcome admitted as a new preparation. It is not a theorem that every physical instrument has this property. Theorem I of the necessity note constructs a more specific two-pointer Hamiltonian protocol even with a prior density ceiling, but has its own explicit access and storage assumptions. In particular a prior cap alone does not supply the postulate needed to preserve a floor after records.

Combining (13) with the cusp preparation-floor theorem shows that its nonempty, finite-covariance preparation class cannot also satisfy the fine-record postulate. Equivalently, with weak closure and affine translations, the records would admit the two point preparations that merge, contradicting its reversible phase-state evolution. This is an incompatibility among assumptions, not a proof that any one of them is physically mandatory.

6. What separates this from the quantum completion?

For every supplied physical $\hbar > 0$, the cusp note's self-adjoint Hamiltonian gives a global quantum unitary group. For a quantum state with finite position and momentum covariances, the Robertson–Schrödinger inequality gives

$$\sqrt{\det \mathrm{Cov}(Q, P)} \geq \hbar/2. \tag{15}$$

This remains true after any selective instrument whose conditional output is a quantum state with finite covariance. There is nevertheless an exact initial-state comparison: set

$$\epsilon_\hbar = \sqrt{\frac{\hbar}{2S_0}}, \qquad \sigma_q = L_0\epsilon_\hbar, \quad \sigma_p = P_0\epsilon_\hbar. \tag{15a}$$

The Gaussian in (5) then has $\kappa = \hbar/2$ and is exactly the Wigner function of the normalized quantum wavefunction

$$\psi_{\hbar,u}(q) = (2\pi\sigma_q^2)^{-1/4} \exp\left[-\frac{(q - u\sigma_q)^2}{4\sigma_q^2}\right]. \tag{15b}$$

Thus covariance saturation and even equality of these *initial* positive phase-space functions do not by themselves select the subsequent quantum propagation or an interference law. The classical limit in Theorem 2 then corresponds to a formal $\hbar \to 0$ family at fixed S_0; no equality of the quantum and classical evolved states is asserted. At fixed physical $\hbar$, a small ratio $\hbar/S_0$ describes a large-characteristic-action regime only when L_0, P_0 specify actual scales of a family of preparations, not merely a change of units.

Equation (15) does not arise by conditioning a positive joint sharp-position–sharp-momentum law: the canonical Weyl operators obey

$$e^{-iaP/\hbar}e^{ibQ/\hbar} = e^{-iab/\hbar}e^{ibQ/\hbar}e^{-iaP/\hbar}, \tag{16}$$

and are noncommuting for general a, b. An approximate joint measurement can still have two classical outcomes; those outcomes are not the nondisturbing sharp joint record assumed in Theorem 4. The positive quantum constant in (15)–(16) is an input. Conversely, the classical history model has no noncommuting observables or demonstrated interference law, and its apparent probabilistic branch splitting in (6) is not a coherent superposition.

There is an independently motivated empirical separator, with additional composite-system hypotheses. Under setting independence and conditional locality, an arbitrary classical hidden state including trajectory memory satisfies $|S_{\mathrm{CHSH}}| \leq 2$; the short proof and its precise assumptions are in the shared-readiness note. A Bell test with spacelike-separated electron-spin measurements reported $S_{\mathrm{CHSH}} = 2.42 \pm 0.20$ in 245 trials (Hensen et al. 2015, primary abstract read). Its setting-independence and locality interpretation is an additional physical premise, motivated by the relativistic causal setup; the recorded statistical uncertainty matters. The normalized inequality is dimensionless, so this observation does not determine an action scale or select quantum theory uniquely. Nonlocal or setting-dependent classical completions are not excluded by this bound.

7. Consequence for STATE

The cusp now distinguishes three logically different requirements: reversibility on complete classical states, weak continuity of evolution on instantaneous phase laws, and preservation of an action floor under selective records. History dynamics satisfies the first with zero action; Theorem 2 obstructs the second at the exact cusp for actual Liouville transport; Theorem 4 shows that unrestricted sharp classical records contradict the third. Theorem 3 derives a conditional positive floor from the actual transport, stability and copying without reversibility,

but its classical entropy-floor witness leaves quantum identification open. A physically justified response modulus or record-disturbance law might remove the classical zero-width family, but neither is established here. The strong claim that Newtonian mechanics requires nonzero $\hbar$ remains open.